Machine Translated by Google

[Chap 4] Representative values of the data

Machine Translated by Google

Technical analysis (descriptive statistics)

- Technical Analysis
  - A methodology for describing or explaining phenomena

- The necessity of technical analysis

  methodologies – Limitations of brain and recognition capacity
  - The ability to see, hear, store in the mind, and perceive trends per unit of time

    The amount of data is extremely limited.

  – A methodology is needed to organize and understand large amounts of data clearly.

- Representative

  methods – Indicative methods (mean, median,

  standard deviation, etc.) – Visualization

Machine Translated by Google

Representative value

• Mean – A value

calculated by adding all data and dividing by the number

of items – Most commonly used, but very vulnerable to extreme values – Ex)

First-year annual salary of an MBA graduate • 70

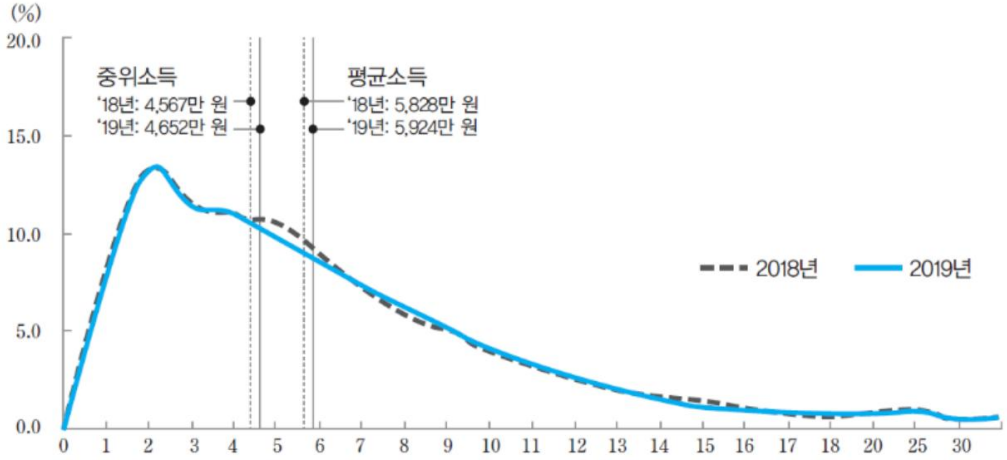
million won, 60 million won, 85 million won, 65 million won, 60 million won, 500 million won

ÿ Average 140 million won (completely unrealistic)

– Example of Average Household

Income • The average household income in Korea is shown to be approximately 60 million won (59.24 million won), but

The fact is that half of the households have an income of 46.52 million won or less



[그림 4-1] 가구소득의 비대칭적 분포 (자료: 통계청 2020년 가계금융복지조사)

Machine Translated by Google

Representative values (continued)

- Median – The value in the

  center of the data – Not significantly

  affected by extreme values or asymmetry of the distribution – Ex) First-

  year annual salary of an MBA graduate • 70 million

  won, 60 million won, 85 million won, 65 million won, 60 million won, 500 million won

  ÿ Median 62.5 million won (much more realistic)

  – The median household income in Korea is 46.52 million won.

  • The discrepancy with reality is much smaller when compared to the average household income of 59.24

  million won. • The median appears lower than the mean because the household income distribution has a long right tail.

- Mode – Setting the most

  frequently occurring value as the representative value

  • In the case of '1, 2, 3, 3, 4, 5', the mode is 3 – The

  mode is useful in categorical data • Example) Identify the most frequently

  occurring word by looking at the frequency of occurrence of each word

Machine Translated by Google

Data variability

- Example) Assume that the average duration of process design methods A, B, and C is 5 minutes – then, can it be concluded that the durations of the three processes are identical?
  - Measure the process time for each of A, B, and C three times: A : (5, 5, 5), B : (4, 5, 6), C : If you obtained (3, 5, 7)?
    - From the perspective of process control, A and C require distinctly different considerations. • The fact that the averages are the same does not mean that the required times for processes A, B, and C are the same.

What makes the three data sets A, B, and C different is the variability of the values.

- The expression 'average is 5 minutes' contains only information about the overall magnitude of the data values, Information on value variability is not included
- Between the magnitude and volatility of the value, cases where volatility has a greater impact on business decision-making

  There are many U

- Indicators expressing volatility
  - Variance and standard deviation are representative – Standard deviation is widely used

Machine Translated by Google

standard deviation

- Standard deviation of process times for A, B, and C

A: (5, 5, 5)

– 평균: $(5+5+5)/3=5$

– 분산: $\{(5-5)^2+(5-5)^2+(5-5)^2\}/(3-1)=0$

– 표준편차: $\sqrt{0}=0$

B: (4, 5, 6)

– 평균: $(4+5+6)/3=5$

– 분산: $\{(4-5)^2+(5-5)^2+(6-5)^2\}/(3-1)=1$

– 표준편차: $\sqrt{1}=1$

C: (3, 5, 7)

– 평균: $(3+5+7)/3=5$

– 분산: $\{(3-5)\ +(5-5)^2+(7-5)^2\}/(3-1)=4$

– 표준편차: $\sqrt{4}=2$

Machine Translated by Google

Data size and variability

- In the previous example, Process A can be considered to have a variation range of approximately 5±0, B 5±1, and C 5±2 – the range obtained by adding or subtracting the standard deviation from the mean is about 70% of the variation range of the corresponding data values. (More precisely, it corresponds to 68%) • 'This value **usually** falls within that range'

  – Adding and subtracting twice the standard deviation (more precisely, 1.96 times) from the mean gives the 95% confidence interval. • 'This value falls **roughly** within that range'

- The three numbers AB and C are not identical! – Numbers with different volatility cannot be treated as identical even if their magnitudes are the same – In the world of data, no matter what value you look at, you must simultaneously consider the front (magnitude) and back (volatility) of the number. Things to consider

Machine Translated by Google

[Python Practice] Extracting Data

- Read data

예제 ex4-1.py (a)

```
import pandas as pd
b = pd.read_csv('buylov.csv')
print(b)
```

- Extraction of specific columns

예제 ex4-1.py (b)

```
import pandas as pd
b = pd.read_csv('buylov.csv')
print(b['BGiftPrice'])
```

Machine Translated by Google

Separating data based on specific column values

- Differentiate data based on Role values
  - Store the data of the gift giver in the variable 'bg' and the data of the recipient in the variable 'br' separately.

예제 ex4-2.py

```
import pandas as pd
b = pd.read_csv('buylov.csv')
bg = b[b['Role']=='Giver']
br = b[b['Role']=='Receiver']
print('< Giver Data >')
print(bg)
print('< Receiver Data >')
print(br)
```

Machine Translated by Google

Calculating the representative value

- Calculating the average price of birthday gifts

예제 ex4-3.py (b)

```
import pandas as pd
import numpy as np
b = pd.read_csv('buylov.csv')
print(b['BGiftPrice'])
print('평균 :', np.mean(b['BGiftPrice']))
```

– The BGiftPrice, or the average price of a birthday gift, was found to be approximately $124.59.

Machine Translated by Google

Calculating the representative value

- Median of birthday gift prices

예제 ex4-3.py (c)

```
import pandas as pd
import numpy as np
b = pd.read_csv('buylov.csv')
bp = b['BGiftPrice']
print(bp)
print('평균 :', np.mean(bp))
print('중앙값 :', np.median(bp))
```

– The average is approximately $124.59, but the median is calculated as $93.0 – The difference between the average and the median is significant.

Considering that the median is not affected by extreme values, one can infer that the average value was skewed toward a higher level due to extremely expensive birthday gifts.

Machine Translated by Google

Calculate standard deviation

- Standard deviation: A representative indicator of value variability

예제 ex4-4.py

```
import pandas as pd
import numpy as np
b = pd.read_csv('buylov.csv')
bp = b['BGiftPrice']
print(bp)
print('평균 :', np.mean(bp))
print('중앙값 :', np.median(bp))
print('표준편차 : ', np.std(bp))
```

– The standard deviation is 114.64, suggesting that the price range of birthday gifts is widely spread. • Confirmed through visualization in Chapter 5.